

Cooker Whistle Detector

Data Collection Requirements & Lessons Learned

1. DATA COLLECTION REQUIREMENTS

1.1 Recording Environment

- Record in the actual kitchen where the product will be used
- Keep the exhaust fan and chimney running during all recordings
- Record with the exact microphone and USB dongle used in the final product
- Use a raw audio capture tool with no noise suppression, AGC, or echo cancellation applied

Class	What to Record	Key Rule
background	• Exhaust fan running alone • Exhaust fan with chimney running together • Gas flame burning with no cooker activity • People speaking near the cooking area • People speaking from a distance • Multiple people talking simultaneously • Sizzling sounds from frying and cooking • Utensils clinking and clattering • Water boiling in an open vessel • TV or radio playing in the background • Complete silence (no fan, no activity) • Minimum 15 minutes across varied conditions	Fan noise must be present
pressure_hiss	• Sustained steam hiss from cooker before whistles • Sustained steam hiss from cooker after whistles • Hiss with exhaust fan running in background • Hiss with kitchen ambient sounds • Different cooker sizes if possible • Minimum 10 minutes	Clips must contain only hiss — no whistle bursts within any clip
cooker_whistle	• Individual whistle bursts from pressure cooker • Minimum 8 minutes	Clips must start and end cleanly — no hiss sections before or after the burst

Note: All three classes can be recorded together in a single continuous session — for example, recording an entire pressure cooker cooking cycle from start to finish. The long recording can then be segmented and labelled into the appropriate classes before uploading to Edge Impulse.

2. HARDWARE & SOFTWARE REQUIREMENTS

2.1 Required Hardware Chain

The following exact hardware chain must be used for both data collection and deployment on the board. Any deviation may cause a train/deploy mismatch.

Component	Specification
Microphone	mic with 3.5mm TRRS jack
USB Adapter	Aropana USB audio dongle (3.5mm to USB) — this is the ADC in the chain
Connection	Mic capsule → 3.5mm TRRS → Aropana dongle → USB port
Sample Rate	44100 Hz (mic native rate — do not let OS resample)
Channels	1 (Mono)
Bit Depth	16-bit PCM

2.2 Recording Software Setup (Windows)

- Software: Audacity v3.7.7 (64-bit installer from GitHub — do not use MuseHub version)
- Audio Host: Windows WASAPI (exclusive mode)
- Recording Device: Microphone (USB Audio Device) — the Arpana dongle
- Channels: 1 (Mono)
- Project Sample Rate: 44100 Hz
- Default Sample Format: 32-bit float during recording is fine

Windows Control Panel check: Sound → Recording → right-click USB mic → Properties → Advanced tab → Default Format: set to '1 channel, 16 bit, 44100 Hz'. Both Exclusive Mode checkboxes must be ticked. If there is no Enhancements tab, the driver has no processing — this is correct.

2.3 Consequences of Changing Hardware

Change	Consequence	Action Required
Different USB dongle (different ADC chip)	Different frequency response — model may generalise poorly for strong signals like whistle/hiss	Retrain with new dongle if accuracy drops
Different mic capsule	Different sensitivity and polar pattern — amplitude levels shift	Retrain with new mic
Phone for recording (any app)	AGC and noise suppression applied at driver level — complete train/deploy mismatch — model fails	Discard all phone recordings. Re-record with correct chain.
Laptop built-in mic	Intel Smart Sound Technology applies beam forming and noise	Do not use under any circumstances

Change	Consequence	Action Required
	suppression — completely different signal	
Same mic + same dongle + different computer	Negligible difference — USB controller noise floor varies but is inaudible to the model	Acceptable — no retraining needed
Windows shared mode (not exclusive)	OS audio enhancements may apply — risky and unverifiable	Always use WASAPI exclusive mode

2.4 Export Format

- File format: WAV (Microsoft)
- Encoding: Signed 16-bit PCM
- Do not export as MP3, FLAC, AAC, or any compressed format — lossy compression destroys frequency content
- In Audacity: File → Export Audio → Export as WAV → Encoding: Signed 16-bit PCM

3. RECORDING WORKFLOW

3.1 Recommended Session Approach

Record one long continuous session covering a full cooker cooking cycle — do not try to start and stop recording for each class. A single 8-10 minute session will cover all three classes.

- Place mic at 1 m from cooker, directly facing it, at counter height
- Start recording in Audacity before turning on the flame
- Note timestamps as events occur: when fan turns on, when hiss starts, when each whistle occurs
- Let the session run continuously until cooking is complete
- Stop recording and export the full session as WAV

3.2 Segmenting the Recording

After recording, open the WAV file in Audacity and use the spectrogram view to identify class boundaries clearly.

- View → Show Spectrum: helps visually locate whistle bursts (sharp vertical lines) and hiss (sustained mid-frequency energy)
- Individual clip length: 1 to 10 seconds per clip for Edge Impulse — do not upload a 10-minute file directly (or you can upload directly and segment it there)
- Whistle clips: cut precisely around each burst — no hiss before or after
- Hiss clips: cut sections with only sustained hiss — no whistle bursts anywhere in the clip
- Background clips: any section with no cooker activity — fan, ambient sounds, silence

3.3 Verification Before Uploading

Before uploading any clip to Edge Impulse, verify it visually in Audacity:

- Open clip → View → Show Spectrum
- Whistle clip: confirm there is exactly one sharp transient burst, clean start and end, no hiss tail
- Hiss clip: confirm flat sustained energy, no sharp transients anywhere in the clip
- Background clip: confirm no whistle or hiss events present
- Discard any clip that fails this check — do not upload ambiguous clips

4. UPLOADING TO EDGE IMPULSE

- Upload via Edge Impulse web uploader or CLI: `edge-impulse-uploader --api-key <key> *.wav`
- Label each clip at upload time: `background`, `pressure_hiss`, or `cooker_whistle`
- Do not batch-upload an unlabelled long session — segment and label first
- Split ratio: 80% training / 20% test — Edge Impulse handles this automatically on upload

5. DATA COLLECTION PROBLEMS AND LESSONS LEARNED

Problem 1 — Recording App Preprocessing

Recording on a phone or any consumer app applies noise suppression, AGC, and echo cancellation automatically. These filters modify the audio before saving — making it sound different from what the deployment microphone captures in raw mode.

Lesson: Always record using a raw audio capture tool with no processing applied. The recording device, software, and hardware chain must match the deployment setup exactly.

Problem 2 — Background Noise Not Representative of Real Environment

Background recordings did not include the exhaust fan running. In real kitchen use, the fan is always on during cooking. The model had never seen fan noise so it misclassified it as a target event.

Lesson: Record background in the actual kitchen environment with all appliances running as they would be during normal cooking — fan, gas flame, everything.

Problem 3 — Hiss Clips Contained Whistle Bursts

Some hiss recordings had whistle bursts within them. The model could not learn clean class boundaries when a single clip contained multiple sound types.

Lesson: Each clip must contain only one sound type. Hiss clips must have no whistle content and whistle clips must have no hiss content.

Problem 4 — Training Data Did Not Match Real Deployment Environment

Initial dataset was recorded in controlled environments. Real kitchen has fan noise, flame sound, sizzle, and room reverb. Model trained on clean data failed completely in real conditions.

Lesson: Training data must be recorded in the actual deployment environment — same kitchen, same background noise, same acoustic conditions as where the product will be used.

Problem 5 — Phone Playback Re-Recording

Audio recorded correctly on the board mic was played back through a phone speaker and re-recorded. This introduced phone speaker frequency response limits, room reverb, and speaker compression — making the spectrogram completely different from direct mic capture.

Lesson: Never record playback of audio through any speaker. Discard all recordings made this way entirely. Record directly from the sound source using the correct hardware chain.

6. ADDITIONAL CONSIDERATIONS FOR FUTURE ROUNDS

Different Cooker Types

A steel cooker and an aluminium cooker whistle at slightly different frequencies and amplitudes. If the product is used with a cooker type not in the training data, accuracy may drop.

Solution: Include recordings from multiple cooker types and sizes — at minimum steel and aluminium, across 2 L, 3 L, 5 L, and 7 L capacities.

Microphone Placement in Final Product

If the microphone is mounted inside a casing in the final product, the acoustic response changes compared to recording with a bare mic. The model trained with a bare mic may not perform the same once enclosed.

Solution: For the first dataset, record at 1 m distance, directly facing the cooker, at counter height. For subsequent rounds, record at 0.5 m, 1 m, 1.5 m, and 2 m, and at 45 degrees to the side.

Very Loud Kitchen Environments

Commercial kitchens or kitchens with unusually loud exhaust systems may have background noise levels that overwhelm the whistle signal.

Solution: Include recordings from at least two or three different kitchen environments — quiet home kitchen, busy home kitchen, and if possible a semi-commercial kitchen.